Alfie McGlennon

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Summary

Early-career climate data scientist completing an MSc in Climate Change and Artificial Intelligence at the University of Reading, with a BSc in Meteorology and Climate.
Experienced in analysing large-scale climate datasets (ERA5, Berkeley Earth) using Python to study extremes, anomalies and long-term trends, and in translating results into clear, decision-focused visualisations.
Seeking roles in climate analytics, climate risk, or applied environmental data science within the Thames Valley / hybrid UK context.

Skills

Core Analytical & Data Skills

- Data analysis & preprocessing
- Time-series analysis
- Exploratory data analysis (EDA)
- Statistical analysis & anomaly detection
- Quantitative modelling

Climate & Environmental Data

- Climate data analysis (gridded datasets)
- Reanalysis & observational datasets (ERA5, Berkeley Earth)
- Climate anomalies, trends & extremes
- Interpretation of large-scale environmental patterns
- Climate risk & impact-relevant analysis

Programming & Tools

- Python (pandas, NumPy, matplotlib, Seaborn, SKLearn)
- Data visualisation & dashboards (Power BI, DAX)
- SQL, Excel
- Git, JupyterLab

Machine Learning & AI (Foundational / Applied)

- Machine learning fundamentals
- Feature engineering
- Supervised learning (tree-based models)
- Model interpretation (e.g. SHAP – applied in project work)
- Ongoing work with XGBoost and neural networks

Projects

Rainfall Extremes Dashboard (Power BI, Python, ERA5) | 2025

• Built an interactive dashboard analysing rainfall variability and extreme-event frequency across regions.
• Processed multi-year ERA5 precipitation data in Python to compute anomalies and regional aggregates.
• Designed visual outputs to communicate hydro-climatic risk to non-technical audiences.

City Climate Bars & Stripes Web App (Python, D3.js) | 2025

• Processed multi-decadal Berkeley Earth temperature datasets for city-level climate visualisation.
• Optimised large CSV datasets for efficient web rendering.
• Integrated Python-based pre-processing with a D3.js frontend to enable intuitive exploration of long-term climate trends at the city scale.

Climate Playbook – Interactive Climate Education & Analytics Platform (Python, Web, ERA5, TypeScript) | 2026

• Designing an interactive climate-education and analytics platform spanning KS1–KS5 and adult learners, grounded in physically realistic representations of the climate system.
• Developing uncertainty-aware visualisations and simplified modelling tools to explore climate variability, extremes, and long-term risk.
• Building data pipelines and learning pathways that introduce users to real climate datasets and workflows, including reanalysis data (e.g. ERA5).
• Translating complex climate science into transparent, accessible interactive tools that support exploration, learning, and early-stage climate coding.

Education

University of Reading - Reading, UK | MSc

Climate Change and Artificial Intelligence -Current Study, 2026

Modules include *Applied Data Science with Python, Artificial Intelligence and Machine Learning,* and *Data Science and Climate Services,* which combine advanced analytics and AI techniques with an understanding of the climate system.

University of Reading - Reading, UK | BSc

Meteorology and Climate, 2025

- Dissertation: Investigating an Equilibrium Approximation for the Bowen Ratio (First Class)
- Key Modules: Data Science and Climate Services, Numerical Modelling, Boundary Layer Meteorology
- Grade: 2:1

The Forest School - Wokingham | A-Levels

Mathematics (A), Geography (A), Physics (B), 2021

The Forest School - Wokingham | GCSEs

2019

I passed 11 GCSE subjects, including Mathematics (8), English Language (8), and Physics (8).

Experience

DATA ANALYST INTERN | 07/2024 - 07/2024

Safestore - Borehamwood

- Analysed customer data to identify customer behaviour trends and retention metrics.
- Delivered concise presentations to senior management using PowerPoint.
- Used Python (pandas, matplotlib, seaborn) to create data visualisations to support business retention strategies.

SWIM TEACHER (LEVEL 2) | 11/2020 - Current

Places Leisure - Reading

- Delivered structured instruction to students of all levels, adapting methods to diverse learning styles.
- Developed clear communication, patience and management skills.

LIFEGUARD | 08/2019 - Current

Places Leisure - Reading

- Worked as part of a team to maintain safety standards, respond to incidents, and support an inclusive environment.